ROUND 1 / AI SYNERGY CASE CHALLENGE

CHOOSE A THEME. INVESTIGATE THE CASE. DEFINE THE PROBLEM.

Each theme is a problem space, not a ready-made problem statement.

Study the evidence, stakeholders, constraints and unknowns. Compare possible problems, justify the one worth solving, and show how Human-AI reasoning led you there.

1. **AI for All**
   When access exists, but meaningful adoption does not.
2. **Agriculture**
   The decision window is short; the information is fragmented.
3. **Healthcare**
   The visit ends, but continuity of care does not.
4. **Tourism**
   Visitors arrive, yet local experiences remain invisible.
5. **Energy & Environment**
   Demand, weather and infrastructure collide.
6. **Sports**
   Performance is visible; potential is harder to measure.
7. **Governance**
   One citizen issue can cross many departments.

HOW ROUND 1 WORKS

CASE
Understand the situation

INVESTIGATE
Question assumptions

DIVERGE
Frame alternatives

DECIDE
Choose one problem

DESIGN
Propose + specify

DOCUMENT
Show the AI trail

YOU ARE NOT CHOOSING A PROBLEM STATEMENT.

You are choosing a problem space.

CASE -> INVESTIGATION -> PROBLEM DISCOVERY -> AI REASONING -> SOLUTION -> BUILD

ROUND 1: CASE INVESTIGATION CHALLENGE

Participant instructions - read before selecting a case

YOUR TASK

Choose ONE case from this booklet. The case is not the problem statement. Your team must investigate the situation, compare plausible problem framings, select one specific problem, and defend why it is worth solving.

This case file intentionally contains ambiguity. Some signals may point in different directions. You are not expected to solve every issue in a case, and there is no single official problem statement. Strong teams will narrow the case into a precise, evidence-supported problem that can be tested and built upon in the next round.

WHAT ROUND 1 SHOULD DEMONSTRATE

- Problem discovery before solution generation.
- Purposeful use of AI for investigation, reasoning, comparison, research support, design and technical exploration.
- Human ownership of consequential decisions - including when AI suggestions are accepted, modified or rejected.
- Clear separation of case facts, assumptions, inferences, unknowns and external evidence.
- A feasible solution scope that can become a working implementation in the on-campus Build Round.

REQUIRED ROUND 1 PACKAGE

- Case Investigation / Problem-Solution Brief (follow the platform instructions for page and file limits).
- Solution Presentation.
- Genuine AI interaction logs or accessible share links, plus an index of consequential decisions.
- Basic SRS / implementation plan with users, inputs, outputs, workflow, functional requirements, AI/human responsibilities, constraints, error handling and acceptance tests.
- GitHub / prototype / notebook / architecture / wireframe / technical proof, if available. A complete product is not required in Round 1.
- References, reused assets declaration and team contribution declaration.

IMPORTANT

A polished deck, large prompt count, paid AI subscription or large amount of code does not independently earn credit. Evaluation focuses on problem diagnosis, evidence, meaningful Human-AI collaboration, feasibility, testability and accountability.

HOW TO WORK WITH THE CASE FILE

A suggested investigation sequence - not a mandatory solution template

1. MAP	List stakeholders, signals, constraints and unknowns before proposing a product.
2. DIVERGE	Generate at least two credible problem framings. Ask what changes if a different stakeholder becomes primary.
3. INVESTIGATE	Use AI and external sources to test assumptions. Record what is case evidence versus outside evidence.
4. DECIDE	Select one problem. Explain why it is more important, tractable or testable than the alternatives you considered.
5. DESIGN	Define the solution workflow, AI role, human role, failure modes, safeguards and minimum viable scope.
6. VERIFY	Specify acceptance tests, baseline comparison, outcome metric and one harm/exclusion metric.
7. DOCUMENT	Preserve genuine AI logs and show the decisions they influenced.

SYNERGY DECISION TRAIL - RECOMMENDED INDEX FORMAT

Decision	Initial view	AI contribution	Evidence / verification	Final human decision
Example decision				
Example decision				
Example decision				

EVIDENCE LABELS

When writing your brief, explicitly distinguish: CASE EVIDENCE - information provided in this booklet; EXTERNAL EVIDENCE - independently sourced information; ASSUMPTION - something you currently believe but have not verified; UNKNOWN - information that could materially change your decision.

01

AI FOR ALL

The Last-Mile AI Gap

Case scenario - your team must define the problem statement

SITUATION

A state-funded college network has introduced generative-AI access for students, but usage is uneven. Some students use AI daily for coding, writing and research; others avoid it because of language barriers, weak devices, low confidence or uncertainty about what is allowed. Faculty responses also vary: some encourage AI with disclosure, while others prohibit it completely. The institution wants to improve AI readiness without turning the programme into a generic tool-training workshop.

STAKEHOLDERS

Undergraduate students across disciplines | Faculty and academic administrators | Students using low-end devices or limited data | Students learning primarily in Hindi or another regional language | Placement and career-development teams

CASE EVIDENCE PACK

Signal	Case evidence	What remains uncertain
Access	78% of surveyed students report having a smartphone; 46% have reliable access to a personal laptop.	Whether device ownership translates into meaningful AI access.
Usage	A simulated survey of 600 students shows 31% use generative AI daily, 37% occasionally, and 32% rarely or never.	Why non-users avoid AI: access, trust, rules, language, or perceived usefulness.
Language	42% say they are more comfortable explaining complex ideas in Hindi or mixed Hindi-English.	Whether multilingual interaction improves learning outcomes.
Policy	Different departments currently communicate different rules for acceptable AI use.	How much policy inconsistency affects behaviour.
Capability	Faculty report that polished AI-assisted work does not always correspond to student understanding.	How to measure genuine understanding without banning AI.
Equity	Students with better devices and paid tools appear more confident in AI-heavy assignments.	Whether tool access is causing a meaningful performance gap.

QUESTIONS THE CASE LEAVES OPEN

- Is the primary problem access, literacy, confidence, assessment design, policy ambiguity, or something else?
- What should be measured: AI usage, AI literacy, learning outcomes, fairness, or employability?
- What level of AI support should remain optional?
- How can the system avoid penalising students who use fewer tools?

BOUNDARY

Do not assume that more AI usage is automatically better. Do not design a surveillance system that reads unrelated private conversations. Any individual capability claim must be based on observable task evidence.

ROUND 1 OUTPUT FOR THIS CASE: Compare plausible problem framings, select one specific problem, justify it with evidence, propose a feasible solution, define the SRS / implementation plan, and preserve the AI reasoning trail that materially shaped your decisions.

02

AGRICULTURE

The Harvest Decision Window

Case scenario - your team must define the problem statement

SITUATION

A cluster of smallholder vegetable farmers must decide when and where to sell a highly perishable crop. Prices differ across nearby mandis, transport is expensive, connectivity is intermittent, and farmers often depend on word-of-mouth information. A sudden weather change can shift harvest timing by a day or two, which can materially affect quality and price. Several digital agriculture apps already exist, but adoption in the cluster is low.

STAKEHOLDERS

Small and marginal farmers | Local aggregators and transporters | Mandi traders and market committees | Agriculture extension workers | Farmer producer organisations

CASE EVIDENCE PACK

Signal	Case evidence	What remains uncertain
Price spread	On the same morning, simulated mandi quotes for the crop range from Rs 14/kg to Rs 24/kg across four markets.	Whether quoted prices are actually obtainable by a specific farmer.
Transport	A 90 km market may offer a higher price, but transport and loading can remove Rs 4-6/kg of the advantage.	How costs vary by volume and shared transport.
Connectivity	Two of five villages report unreliable mobile data during parts of the day.	Whether SMS, IVR, WhatsApp, offline-first or human intermediaries are better.
Trust	Farmers say they trust local traders and neighbours more than unfamiliar app recommendations.	What evidence would be needed to change or supplement trusted routines.
Weather	Forecasts indicate possible heavy rain within 36 hours.	How local conditions differ from district-level forecasts.
Waste	Farmers estimate 8-15% post-harvest loss in weeks of oversupply.	Which part of the loss is avoidable through better decisions.

QUESTIONS THE CASE LEAVES OPEN

- Is the highest-value intervention about price discovery, logistics, harvest timing, collective selling, trust, or something else?
- What information must be real-time and what can be approximate?
- Who bears the cost of incorrect advice?
- How would your approach work during poor connectivity?

BOUNDARY

Do not claim a guaranteed price, yield or income improvement. Any AI-generated recommendation must expose uncertainty and should not replace authorised agronomic or financial advice where such advice is required.

ROUND 1 OUTPUT FOR THIS CASE: Compare plausible problem framings, select one specific problem, justify it with evidence, propose a feasible solution, define the SRS / implementation plan, and preserve the AI reasoning trail that materially shaped your decisions.

03

HEALTHCARE

The Follow-Up Blind Spot

Case scenario - your team must define the problem statement

SITUATION

A district hospital runs high-volume outpatient clinics. Clinicians can identify patients who need follow-up, but the system struggles to ensure that those patients actually return at the right time with the right information. Phone numbers may be shared across families, some patients travel long distances, and staff have limited time for manual calling. The hospital wants to reduce avoidable loss-to-follow-up without introducing unsafe automated medical decisions.

STAKEHOLDERS

Patients and caregivers | Doctors and nurses | Registration and records staff | Community health workers | Hospital administration

CASE EVIDENCE PACK

Signal	Case evidence	What remains uncertain
Volume	The simulated clinic sees about 900 outpatient visits per day.	How many genuinely require active follow-up.
Return rate	In a three-month sample, 28% of patients marked for follow-up did not return within the recommended window.	Which missed visits are clinically important versus administratively delayed.
Contact	19% of records contain a shared household phone number; 7% have an unreachable number.	What alternative contact channels are acceptable and consented.
Records	Follow-up instructions are written as free text and vary by clinician.	Whether structured capture would reduce ambiguity.
Staffing	A small records team cannot manually prioritise every missed appointment.	What prioritisation can be safely automated.
Risk	Clinical teams reject any system that independently diagnoses, changes treatment, or gives emergency advice.	Where the safe boundary between workflow support and clinical judgement lies.

QUESTIONS THE CASE LEAVES OPEN

- Is the problem reminders, prioritisation, information quality, transport/access, continuity of care, or another bottleneck?
- What information can safely determine urgency?
- How will consent and privacy be handled?
- How will the system respond to missing or conflicting information?

BOUNDARY

This is a workflow and continuity-of-care case, not a diagnostic challenge. Do not propose autonomous diagnosis, treatment changes or emergency triage. Human clinical review must remain authoritative.

ROUND 1 OUTPUT FOR THIS CASE: Compare plausible problem framings, select one specific problem, justify it with evidence, propose a feasible solution, define the SRS / implementation plan, and preserve the AI reasoning trail that materially shaped your decisions.

04

TOURISM

Beyond the Fort

Case scenario - your team must define the problem statement

SITUATION

A heritage city receives strong visitor traffic at a small number of famous landmarks, while many nearby cultural sites, craft communities, food businesses and local experiences remain poorly discovered. Visitors have limited time, information is fragmented across websites and social media, and smaller operators struggle to be visible. At the same time, residents are concerned that indiscriminate promotion could create congestion, misinformation or disrespectful behaviour at sensitive sites.

STAKEHOLDERS

Domestic and international visitors | Local guides and tourism officials | Craftspeople and small businesses | Residents near heritage sites | Transport providers and heritage managers

CASE EVIDENCE PACK

Signal	Case evidence	What remains uncertain
Concentration	Simulated visitor counts show 67% of tracked tourist visits concentrated at two major attractions.	Whether concentration is due to preference, discoverability, time, transport or trust.
Stay length	A majority of surveyed day visitors report having 5-8 hours in the city.	How much itinerary complexity is realistic.
Discovery	Small operators report depending heavily on word-of-mouth and social media.	How to verify listings and keep information current.
Mobility	Several lesser-known sites are 3-8 km from the main visitor corridor.	Whether route, transport, safety or accessibility is the main barrier.
Authenticity	Local stakeholders complain that online descriptions frequently copy outdated or inaccurate facts.	How authoritative local information can be maintained.
Sensitivity	Some locations have religious, conservation or community-specific norms.	How recommendations should respect restrictions and local context.

QUESTIONS THE CASE LEAVES OPEN

- Is the key problem discovery, trust, itinerary planning, mobility, local economic inclusion, content quality, or something else?
- Whose definition of a good tourism outcome should dominate?
- How should the system behave when data is stale?
- How can promotion avoid overwhelming fragile locations?

BOUNDARY

Do not invent historical facts, opening hours or accessibility claims. Clearly distinguish verified information, stakeholder-provided information and inferred recommendations.

ROUND 1 OUTPUT FOR THIS CASE: Compare plausible problem framings, select one specific problem, justify it with evidence, propose a feasible solution, define the SRS / implementation plan, and preserve the AI reasoning trail that materially shaped your decisions.

05

ENERGY & ENVIRONMENT

The 4 PM Spike

Case scenario - your team must define the problem statement

SITUATION

A large educational campus experiences recurring electricity-demand spikes in the late afternoon and evening. It has rooftop solar, multiple academic buildings, hostels, labs and cooling loads, but energy data comes from different meters and is not consistently interpreted. Facilities staff want to reduce avoidable peak demand and improve sustainability without disrupting teaching, research or student welfare.

STAKEHOLDERS

Campus facilities and electrical teams | Students and hostel residents | Laboratories and academic departments | Administration and finance teams | Sustainability or environment committees

CASE EVIDENCE PACK

Signal	Case evidence	What remains uncertain
Peak	Simulated meter data shows weekday demand rising sharply between 4 PM and 8 PM.	Which loads cause the spike and which are flexible.
Solar	Rooftop solar output drops as evening demand begins to rise.	Whether storage, load shifting or behavioural changes are economically sensible.
Cooling	Cooling accounts for a large share of afternoon consumption in warm months.	How much can be changed without affecting comfort or equipment.
Meters	Some buildings have 15-minute data; others only monthly totals.	How to operate with uneven data quality.
Critical loads	Labs and server rooms include equipment that cannot be interrupted.	How to distinguish flexible from non-flexible loads.
Behaviour	Past awareness campaigns produced a short-lived reduction in consumption.	Whether feedback, incentives, automation or policy would sustain change.

QUESTIONS THE CASE LEAVES OPEN

- Is the problem prediction, scheduling, visibility, incentives, equipment control, data quality, or another issue?
- What is the safest minimum viable intervention?
- Which loads should never be automated?
- What baseline will prove the intervention actually helps?

BOUNDARY

Do not assume direct control over electrical infrastructure unless your design includes appropriate authorisation and safety controls. Simulated control systems must be labelled as simulations.

ROUND 1 OUTPUT FOR THIS CASE: Compare plausible problem framings, select one specific problem, justify it with evidence, propose a feasible solution, define the SRS / implementation plan, and preserve the AI reasoning trail that materially shaped your decisions.

06

SPORTS

Selection Under Pressure

Case scenario - your team must define the problem statement

SITUATION

A university sports programme conducts open trials for several team sports. Coaches must evaluate many students in limited time, and participants believe selection can favour athletes who already have visibility or prior coaching. Performance data is inconsistent across sports, while coaches argue that numbers alone cannot capture tactical awareness, teamwork or match context. The programme wants a fairer and more transparent selection process without replacing coach judgement with an opaque algorithm.

STAKEHOLDERS

Student athletes | Coaches and selectors | Sports administration | Strength and conditioning staff | Students from less-resourced sporting backgrounds

CASE EVIDENCE PACK

Signal	Case evidence	What remains uncertain
Scale	More than 300 students may attend trials across multiple sports in one week.	Whether the bottleneck is observation time, data capture or decision consistency.
Metrics	Some sports have measurable drills; others depend heavily on contextual judgement.	Which measures are comparable and which should remain sport-specific.
Bias concern	Participants report uncertainty about why they were or were not shortlisted.	Whether transparency would improve trust even if final selections remain similar.
Video	Phones can capture trial footage, but camera angles and lighting vary.	Whether computer vision would be reliable enough for the intended use.
Opportunity	Some students have years of formal coaching; others are talented but inexperienced.	How to avoid equating prior access with potential.
Privacy	Participants may not want footage or performance data reused outside selection.	What consent, retention and deletion rules are needed.

QUESTIONS THE CASE LEAVES OPEN

- Is the problem fairness, consistency, feedback, talent discovery, documentation, or something else?
- Which decisions should AI support versus never make?
- How will different sports be handled without false comparability?
- What appeal or review process should exist?

BOUNDARY

Do not produce medical injury predictions from limited trial data. Do not design an autonomous final selector. Any video or biometric-like analysis requires explicit consent and clear retention limits.

ROUND 1 OUTPUT FOR THIS CASE: Compare plausible problem framings, select one specific problem, justify it with evidence, propose a feasible solution, define the SRS / implementation plan, and preserve the AI reasoning trail that materially shaped your decisions.

07

GOVERNANCE

One Complaint, Five Departments

Case scenario - your team must define the problem statement

SITUATION

A municipal grievance system receives complaints through a portal, phone calls, social media and in-person offices. The same issue may be reported multiple times using different wording, and residents often do not know which department is responsible. Staff spend time forwarding cases, while citizens receive generic status updates that do not explain what is happening. The municipality wants to improve responsiveness without allowing an automated system to silently close or deprioritise legitimate complaints.

STAKEHOLDERS

Residents and complainants | Municipal grievance staff | Engineering, sanitation, water and other departments | Ward-level officials | Senior administrators and auditors

CASE EVIDENCE PACK

Signal	Case evidence	What remains uncertain
Channels	Complaints arrive through at least four different channels and are stored in partially separate workflows.	How many records are duplicates versus distinct incidents.
Routing	A simulated sample shows 23% of grievances were reassigned at least once.	Whether reassignment reflects unclear categories or organisational responsibility.
Language	Residents submit complaints in English, Hindi and mixed-language text; some include photos or voice notes.	How reliably complaints can be interpreted without losing nuance.
Status	Citizens frequently see generic labels such as "in process" for several days.	What level of explanation can be safely exposed.
Urgency	Some low-frequency complaints may involve high public-safety importance.	How to avoid popularity or volume becoming the only priority signal.
Accountability	Officials require an auditable record of why a grievance was routed or prioritised.	What explanations are meaningful and reviewable.

QUESTIONS THE CASE LEAVES OPEN

- Is the most important problem intake, deduplication, routing, prioritisation, transparency, accountability, or another failure?
- How should AI handle sarcasm, incomplete descriptions or conflicting reports?
- What decisions require human confirmation?
- How can a resident challenge an incorrect classification?

BOUNDARY

Do not design a system that automatically rejects or closes citizen complaints solely on an AI score. Preserve human review, auditability and an accessible correction or appeal path.

ROUND 1 OUTPUT FOR THIS CASE: Compare plausible problem framings, select one specific problem, justify it with evidence, propose a feasible solution, define the SRS / implementation plan, and preserve the AI reasoning trail that materially shaped your decisions.

FINAL CHECK BEFORE YOU SUBMIT

Use this checklist to make sure your Round 1 entry is judgeable and verifiable.

☐	We selected exactly one case and clearly identified the stakeholder(s) we are designing for.
☐	We considered at least two plausible problem framings before selecting our final problem.
☐	Our final problem statement describes a specific failure, unmet need or uncertainty - not merely a technology we want to build.
☐	We marked assumptions and unknowns instead of presenting them as facts.
☐	Important external claims are supported by sources.
☐	Our AI logs are genuine and preserve meaningful changes of direction, not retrospectively rewritten conversations.
☐	Our decision index points judges to the AI interactions that influenced consequential decisions.
☐	Our proposed solution clearly separates AI responsibilities from human responsibilities.
☐	We specify what happens when information is missing, conflicting or unreliable.
☐	We include at least three acceptance tests and explain what Version 1 will intentionally not do.
☐	Any prototype, dataset, mock integration or pre-existing component is clearly declared.
☐	We can personally explain every major choice in our submission during an ownership check.

CASE-FILE INTEGRITY NOTE

The numerical entries and operational details in these case scenarios are competition materials created to support problem investigation. Unless a case explicitly identifies a public external source, do not present them as verified real-world statistics about a named organisation or community. You may bring in independent external evidence, but label and cite it clearly.

Shortlisted teams will move from THINK to BUILD in the on-campus round. Your Round 1 proposal should therefore be ambitious enough to matter, but scoped enough that a meaningful working slice can be implemented, tested and demonstrated.

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YOU ARE NOT BEING ASKED

TO GUESS THE ANSWER.

You are being asked to show how you investigate, reason, decide and design with AI - while keeping human ownership.

ROUND 1

FIND THE PROBLEM.
DEFEND YOUR REASONING.

THINK WITH AI.
BUILD BEYOND IT.